PRE-ALGEBRA (7TH) · UNIT 5: OPERATIONS WITH FRACTIONS

Birdhouse Builders

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students build birdhouses in woodshop. Every cut uses mixed-number addition and subtraction, scaling uses multiplication, counting pieces uses division, and design uses fraction equations and customary units.

The saw-cut allowance makes students think carefully about the meaning of each operation.

Standards Addressed

- 7.NS.A.1–3 Operations with rational numbers
- 7.NS.A.2d Convert rational numbers to decimals
- 7.EE.B.4 Equations with fractions
- Customary unit conversions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a real birdhouse or picture. Part 1 measurements and cuts.
Day 2	Part 2 multiple birdhouses and units.
Day 3	Part 3 equations and design.

Materials

- Student handout and Cut List
- Rulers marked in eighths
- Optional: cardboard to build a model
- Calculators (Honors)

How the Three Versions Differ

Level	What changes
CP2	Ordering, adding mixed numbers, simple division of fractions, and feet-inch conversions.
CP1	Saw-cut allowances, scaling, area of fractional rectangles, and one-step fraction equations.
Honors	Cutting optimization across boards, waste fractions, decimals to fractions, powers of fractions, and scale factors.

Watch For (Common Misconceptions)

- Adding numerators and denominators.
- Forgetting to regroup when subtracting mixed numbers.
- Rounding up the number of pieces that fit.

Extension Ideas

- Build a cardboard model from the cut list.
- Research bird species and their entrance hole sizes.
- Find the cost of wood at a local store.

Answer Key – CP2

Part 1: Read the Measurements

#	Task	Answer
1	Order the pieces from shortest to longest.	4, $5\frac{3}{4}$, $6\frac{1}{2}$, $6\frac{3}{4}$, $7\frac{1}{4}$, $9\frac{1}{2}$
2	Write $7\frac{1}{4}$ and $6\frac{3}{4}$ as decimals.	7.25 and 6.75
3	Which is longer, Roof A ($6\frac{1}{2}$ in) or a side ($6\frac{3}{4}$ in)? By how much?	Side, by $\frac{1}{4}$ in
4	Write $9\frac{1}{2}$ as an improper fraction.	$\frac{19}{2}$

Part 2: Plan the Cuts

#	Task	Answer
5	Find the total length of the front and back.	$7\frac{1}{4} + 9\frac{1}{2} = 16\frac{3}{4}$ in
6	Find the total length of all seven pieces (two sides).	$7\frac{1}{4} + 9\frac{1}{2} + 6\frac{3}{4} + 6\frac{3}{4} + 4 + 6\frac{1}{2} + 5\frac{3}{4} = 46\frac{1}{2}$ in
7	A board is 72 inches long. How much is left after cutting one birdhouse (ignore the saw cuts)?	$72 - 46\frac{1}{2} = 25\frac{1}{2}$ in
8	How many $6\frac{3}{4}$-inch sides can you cut from $25\frac{1}{2}$ inches?	$25\frac{1}{2} \div 6\frac{3}{4} = \frac{51}{2} \times \frac{4}{27} \approx 3.8 \rightarrow 3$ sides

Part 3: Customary Units

#	Task	Answer
9	How many feet is $46\frac{1}{2}$ inches?	$46\frac{1}{2} \div 12 = 3\frac{7}{8}$ ft
10	Paint covers $2\frac{1}{2}$ birdhouses per can. How many cans for 10 birdhouses?	$10 \div 2\frac{1}{2} = 4$ cans
11	Draw your own birdhouse and label every piece with a fraction measurement. Find the total length of wood.	Answers vary; check mixed-number addition.

Reflection

#	Prompt	Look for
R1	How do you add mixed numbers with different denominators?	Rewrite the fractions with a common denominator, add whole numbers and fractions, then regroup.
R2	Why do you round DOWN when finding how many pieces you can cut?	A partial piece is too short to use.

Answer Key – CP1

Part 1: Cut Planning

#	Task	Answer
1	Find the total length of the seven pieces.	$46\frac{1}{2}$ in
2	Cutting seven pieces from one board makes 7 cuts. Find the total wood used including saw cuts.	$46\frac{1}{2} + 7(\frac{1}{8}) = 46\frac{1}{2} + \frac{7}{8} = 47\frac{3}{8}$ in
3	How much of the 6 ft board is left? Is it enough for a second front and back?	$72 - 47\frac{3}{8} = 24\frac{5}{8}$ in; front + back + 2 cuts = $16\frac{3}{4} + \frac{1}{4} = 17$ in, so yes.
4	What fraction of the board is wasted as sawdust for one birdhouse?	$\frac{7}{8} \div 72 = \frac{7}{576}$ of the board

Part 2: Many Birdhouses

#	Task	Answer
5	How many inches of wood (with saw cuts) are needed for 12 birdhouses? How many feet?	$12 \times 47\frac{3}{8} = 568\frac{1}{2}$ in = $47\frac{3}{8}$ ft
6	If each 6-ft board is used for one birdhouse, how many boards are needed, and how much wood is left over in all?	12 boards; $12 \times 24\frac{5}{8} = 295\frac{1}{2}$ in (about 24.6 ft) left over.
7	Each birdhouse needs $\frac{3}{4}$ oz of glue. A bottle holds 4 oz. How many bottles?	$12 \times \frac{3}{4} = 9$ oz $\div 4 = 2\frac{1}{4} \rightarrow 3$ bottles
8	Each roof is $6\frac{1}{2}$ in by $5\frac{1}{2}$ in. What is the area of one roof piece?	$6\frac{1}{2} \times 5\frac{1}{2} = \frac{13}{2} \times \frac{11}{2} = \frac{143}{4} = 35\frac{3}{4}$ sq in

Part 3: Equations & Design

#	Task	Answer
9	A back piece plus a perch of length p is $11\frac{1}{4}$ in total. Find p.	$9\frac{1}{2} + p = 11\frac{1}{4} \rightarrow p = 1\frac{3}{4}$ in
10	$\frac{2}{3}$ of a board is 48 in. How long is the board?	$\frac{2}{3}b = 48 \rightarrow b = 72$ in

#	Task	Answer
11	Design a bigger birdhouse by multiplying every length by $1\frac{1}{2}$. Find the new back and side lengths and the total wood.	Back $14\frac{1}{4}$, side $10\frac{1}{8}$; total $46\frac{1}{2} \times 1\frac{1}{2} = 69\frac{3}{4}$ in

Reflection

#	Prompt	Look for
R1	Why does multiplying by $1\frac{1}{2}$ make every piece longer but multiplying by $\frac{2}{3}$ make it shorter?	Multiplying by a number greater than 1 increases a length; by a number less than 1 decreases it.
R2	How did the $\frac{1}{8}$-inch saw cuts change your plan?	They add up, so the total wood is more than the piece lengths alone.

Answer Key – Honors

Part 1: Efficient Cutting

#	Task	Answer
1	Find the length used by one birdhouse with 7 cuts and the leftover from one board.	$47\frac{3}{8}$ in used; $24\frac{5}{8}$ in left
2	Can two leftovers ($24\frac{5}{8}$ in each) make the pieces of a third birdhouse if pieces cannot be joined? Show a cutting plan.	Board 1 leftover: back $9\frac{1}{2}$ + front $7\frac{1}{4}$ + roof B $5\frac{3}{4}$ + 3 cuts = $22\frac{7}{8} \le 24\frac{5}{8}$. Board 2 leftover: 2 sides $13\frac{1}{2}$ + floor 4 + roof A $6\frac{1}{2}$ + 4 cuts = $24\frac{1}{2} \le 24\frac{5}{8}$. Yes.
3	Using that plan, how many boards are needed for 12 birdhouses?	Every 2 boards make 3 birdhouses → 8 boards
4	What fraction of the wood is wasted with the 8-board plan?	Used: $12 \times 46\frac{1}{2} = 558$ in; bought 576 in; waste 18 in = $\frac{1}{32}$

Part 2: Rational Numbers in the Shop

#	Task	Answer
5	A digital caliper reads 0.3125 in. Write it as a fraction in simplest form. Is it closer to $\frac{1}{4}$ or $\frac{3}{8}$?	$\frac{5}{16}$; it is exactly halfway between $\frac{1}{4}$ ($\frac{4}{16}$) and $\frac{3}{8}$ ($\frac{6}{16}$).
6	The entrance hole has diameter $1\frac{1}{4}$ in. Find its area using $\pi \approx \frac{22}{7}$.	$r = \frac{5}{8}$; $A = \frac{22}{7} \times (\frac{5}{8})^2 = \frac{22}{7} \times \frac{25}{64} = \frac{275}{224} \approx 1.23$ sq in
7	Evaluate $(\frac{3}{2})^3$ and explain what it means for a birdhouse scaled by $1\frac{1}{2}$ in every direction.	$\frac{27}{8} = 3\frac{3}{8}$; the volume becomes $3\frac{3}{8}$ times as large.
8	Order $-\frac{3}{8}$, -0.4, $-\frac{5}{16}$, and $-\frac{1}{3}$ (these are errors in inches from four cuts). Which cut was most accurate?	$-0.4 < -\frac{3}{8} < -\frac{1}{3} < -\frac{5}{16}$; $-\frac{5}{16}$ is closest to 0, so it was most accurate.

Part 3: Design Challenge

#	Task	Answer
9	A bluebird needs a floor with area at least 16 sq in. If the floor is square, what side length (to the nearest $\frac{1}{8}$ in) is needed?	√16 = 4 in exactly
10	A flicker needs a floor of $7\frac{1}{4}$ in by $7\frac{1}{4}$ in. By what factor must you scale the original 4 in floor? Solve 4k = $7\frac{1}{4}$.	k = $\frac{29}{16}$ = $1\frac{13}{16}$
11	Write a full cut list for your scaled design and determine how many boards are needed for 6 birdhouses with the least waste.	Answers vary; check scaled lengths, saw cuts, and board plan.

Reflection

#	Prompt	Look for
R1	How did planning leftovers reduce the number of boards?	Combining leftover pieces from two boards produced an extra birdhouse, saving one board in every three birdhouses.
R2	Why does scaling every length by k scale area by k^2 and volume by k^3?	Area multiplies two lengths and volume multiplies three, so the factor is applied that many times.